SAP HANA Operations Guide

How to Release FREE Log Segments in a High Isolation HA DR Site

Controlled recovery procedure for a full log volume on the disaster recovery site

Platform	SAP HANA system replication
Procedure type	Operational remediation
Audience	SAP HANA database administrators
Data handling	Customer identifiers replaced with placeholders

This guide explains how to identify and release log segments that are already in the FREE state on a disaster recovery site when the log filesystem is full. Perform the procedure only under an approved incident or change record and with an administrator who understands the affected SAP HANA topology.

> **Caution** The release command must target only the host and service ports returned for segments in the FREE state. If replication is unhealthy, the site role is unclear, or the result does not match the expected DR system, stop and escalate before making changes.

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Purpose and expected result

The procedure frees filesystem capacity occupied by reusable log segments on the DR site. A successful run reports the number of released segments, reduces the number of FREE entries for the targeted service, and increases available capacity on the log filesystem.

Scope

- Use this procedure for a high isolation HA DR environment in which the DR log filesystem has reached a critical threshold or 100 percent utilization.
- Run the SQL inspection from the primary site and the release operation from the DR operating system.
- Use only site, host, and port values collected from the current environment. Do not reuse values from a previous incident.

Required access and information

Requirement	Why it is needed	Evidence to capture
Approved incident or change	Confirms authorization and scope	Ticket number and approver
Primary SYSTEMDB access	Reads replication and log segment views	DR site name plus query results
DR operating system access as <SID>adm	Starts the HANA nameserver console	Target host and login context
Current host and service ports	Targets only services with FREE segments	Unique <HOST> and <PORT> pairs

Tip Open a secure working note for the incident. Record the site name, host and port pairs, segment counts, and filesystem usage before and after the change. Do not place customer identifiers in shared documentation.

Placeholder reference

Placeholder	Replace with
`<DR_SITE_NAME>`	Exact secondary site name shown in replication status
`<DR_HOST>`	Host returned by M_LOG_SEGMENTS for FREE entries
`<SERVICE_PORT>`	Service port returned for the selected host
`<SID>adm`	Operating system administrator account for the SAP system

Procedure overview

Step	Location	Action	Success indicator
1	DR operating system	Confirm the log filesystem is full	Usage and mount point recorded
2	Primary SYSTEMDB	Confirm replication and obtain the DR site name	Exact site name recorded
3	Primary SQL console	Find FREE segments by host and port	Target pairs recorded
4	DR operating system	Open the nameserver console	hdbcons returns OK
5	Nameserver console	Release FREE segments for each target pair	Released segment count returned
6	Both sites	Verify space and replication	Capacity improves and replication remains healthy

1 Confirm the filesystem condition

Log on to the DR operating system and confirm that the affected mount point is the SAP HANA log filesystem. Record the current size, used space, available space, and utilization percentage.

```
df -h
# Confirm the row for /hana/log or the configured HANA log mount point
```

Caution Do not assume that the largest filesystem is the affected log volume. Confirm the mount point used by the target SAP HANA system before continuing.

Tip Save the command output in the incident record. This creates a baseline for the final verification and helps distinguish released capacity from unrelated filesystem changes.

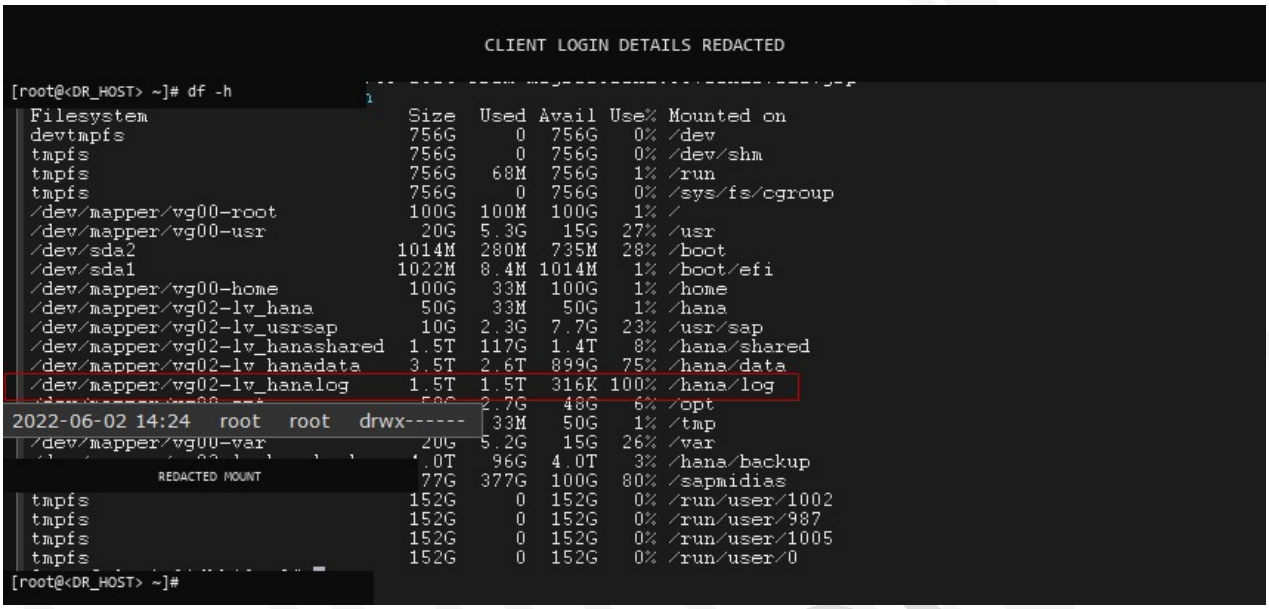


Figure 1 Log filesystem at 100 percent utilization before the release

2 Identify the DR site

Connect to the primary SYSTEMDB in SAP HANA Studio or SAP HANA cockpit. Open the system replication view and verify the current replication topology. Record the exact DR site name.

1. Open the primary SYSTEMDB administration view.
2. Navigate to the system replication status page.
3. Confirm the intended secondary site and copy its site name exactly as displayed.

Caution If the replication status shows an error, an unexpected role, or a different secondary site than the incident scope, stop. Resolve the topology question before running the release command.

Tip Site names are case sensitive inside the quoted schema name. Copy the value instead of typing it from memory.

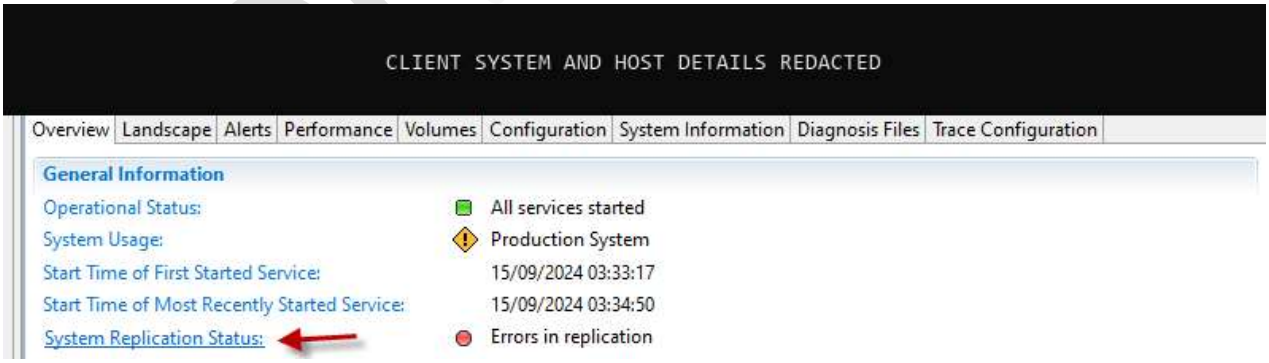


Figure 2 SYSTEMDB overview and system replication status link